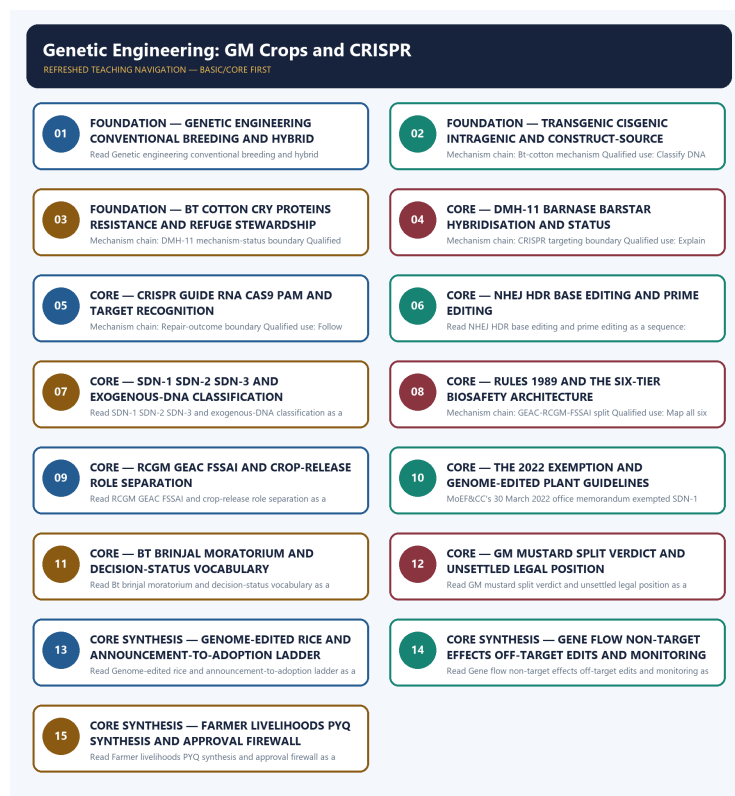


SOLVED PRACTICE WORKBOOK

Genetic Engineering: GM Crops and CRISPR - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Engineering-breeding boundary?

A. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. B. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. C. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. D. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship.

Answer: A. Explanation: Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Engineering-breeding boundary?

A. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. B. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. C. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. D. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation.

Answer: B. Explanation: Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Engineering-breeding boundary without changing its institution, unit or status?

A. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. B. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. C. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. D. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit.

Answer: C. Explanation: Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Engineering-breeding boundary?

A. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. B. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. C. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. D. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic.

Answer: D. Explanation: Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Transgenic-cisgenic-intragenic boundary?

A. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. B. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. C. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. D. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation.

Answer: A. Explanation: Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Transgenic-cisgenic-intragenic boundary?

A. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. B. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. C. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. D. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting.

Answer: B. Explanation: Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Transgenic-cisgenic-intragenic boundary without changing its institution, unit or status?

A. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. B. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. C. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. D. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ.

Answer: C. Explanation: Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Transgenic-cisgenic-intragenic boundary?

A. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. B. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. C. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. D. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged.

Answer: D. Explanation: Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Bt-cotton mechanism?

A. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. B. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. C. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. D. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing,

not established commercial cultivation.

Answer: A. Explanation: Bt cotton expresses cry proteins derived from Bacillus thuringiensis against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Bt-cotton mechanism?

A. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. B. Bt cotton expresses cry proteins derived from Bacillus thuringiensis against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. C. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. D. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting.

Answer: B. Explanation: Bt cotton expresses cry proteins derived from Bacillus thuringiensis against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Bt-cotton mechanism without changing its institution, unit or status?

A. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. B. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. C. Bt cotton expresses cry proteins derived from Bacillus thuringiensis against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. D. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption.

Answer: C. Explanation: Bt cotton expresses cry proteins derived from Bacillus thuringiensis against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Bt-cotton mechanism?

A. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. B. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. C. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. D. Bt cotton expresses cry proteins derived from Bacillus thuringiensis against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship.

Answer: D. Explanation: Bt cotton expresses cry proteins derived from Bacillus thuringiensis against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies DMH-11 mechanism-status boundary?

A. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. B. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. C. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. D. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ.

Answer: A. Explanation: DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of DMH-11 mechanism-status boundary?

A. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. B. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. C. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. D. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit.

Answer: B. Explanation: DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses DMH-11 mechanism-status boundary without changing its institution, unit or status?

A. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. B. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. C. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. D. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption.

Answer: C. Explanation: DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about DMH-11 mechanism-status boundary?

A. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. B. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. C. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. D. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation.

Answer: D. Explanation: DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies CRISPR targeting boundary?

A. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. B. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. C. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. D. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ.

Answer: A. Explanation: In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of CRISPR targeting boundary?

A. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. B. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. C. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. D. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels.

Answer: B. Explanation: In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses CRISPR targeting boundary without changing its institution, unit or status?

A. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. B. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. C. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. D. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels.

Answer: C. Explanation: In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about CRISPR targeting boundary?

A. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. B. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. C. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. D. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting.

Answer: D. Explanation: In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Repair-outcome boundary?

A. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. B. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. C. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. D. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ.

Answer: A. Explanation: A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Repair-outcome boundary?

A. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. B. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. C. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. D. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others.

Answer: B. Explanation: A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Repair-outcome boundary without changing its institution, unit or status?

A. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. B. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. C. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. D. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate.

Answer: C. Explanation: A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Repair-outcome boundary?

A. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. B. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. C. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. D. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit.

Answer: D. Explanation: A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Base-prime editing boundary?

A. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. B. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. C. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. D. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels.

Answer: A. Explanation: Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Base-prime editing boundary?

A. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. B. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. C. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. D. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels.

Answer: B. Explanation: Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Base-prime editing boundary without changing its institution, unit or status?

A. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. B. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. C. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. D. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks.

Answer: C. Explanation: Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Base-prime editing boundary?

A. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. B. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. C. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. D. Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ.

Answer: D. Explanation: Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies SDN-category boundary?

A. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. B. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. C. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. D. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate.

Answer: A. Explanation: SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of SDN-category boundary?

A. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. B. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. C. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. D. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks.

Answer: B. Explanation: SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses SDN-category boundary without changing its institution, unit or status?

A. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. B. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. C. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. D. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks.

Answer: C. Explanation: SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about SDN-category boundary?

A. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. B. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. C. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. D. SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption.

Answer: D. Explanation: SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Six-tier biosafety architecture?

A. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. B. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. C. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. D. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate.

Answer: A. Explanation: The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Six-tier biosafety architecture?

A. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. B. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. C. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. D. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks.

Answer: B. Explanation: The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Six-tier biosafety architecture without changing its institution, unit or status?

A. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. B. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. C. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. D. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed.

Answer: C. Explanation: The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Six-tier biosafety architecture?

A. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. B. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. C. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. D. The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels.

Answer: D. Explanation: The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies GEAC-RCGM-FSSAI split?

A. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. B. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. C. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. D. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status.

Answer: A. Explanation: RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of GEAC-RCGM-FSSAI split?

A. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. B. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. C. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. D. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed.

Answer: B. Explanation: RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses GEAC-RCGM-FSSAI split without changing its institution, unit or status?

A. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. B. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. C. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. D. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes.

Answer: C. Explanation: RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about GEAC-RCGM-FSSAI split?

A. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. B. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. C. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. D. RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others.

Answer: D. Explanation: RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Rules-1989 boundary?

A. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. B. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. C. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. D. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks.

Answer: A. Explanation: The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Rules-1989 boundary?

A. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. B. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. C. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. D. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed.

Answer: B. Explanation: The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Rules-1989 boundary without changing its institution, unit or status?

A. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. B. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. C. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. D. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed.

Answer: C. Explanation: The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Rules-1989 boundary?

A. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. B. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. C. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. D. The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate.

Answer: D. Explanation: The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies 2022 exemption-guideline boundary?

A. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. B. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. C. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. D. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status.

Answer: A. Explanation: MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of 2022 exemption-guideline boundary?

A. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. B. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. C. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. D. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed.

Answer: B. Explanation: MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses 2022 exemption-guideline boundary without changing its institution, unit or status?

A. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. B. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. C. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. D. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically.

Answer: C. Explanation: MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about 2022 exemption-guideline boundary?

A. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. B. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. C. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. D. MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks.

Answer: D. Explanation: MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Bt-brinjal moratorium boundary?

- A. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status.
- B. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary.
- C. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed.
- D. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes.

Answer: A. Explanation: The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Bt-brinjal moratorium boundary?

- A. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes.
- B. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status.
- C. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary.
- D. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically.

Answer: B. Explanation: The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Bt-brinjal moratorium boundary without changing its institution, unit or status?

- A. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically.
- B. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary.
- C. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status.
- D. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing.

Answer: C. Explanation: The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Bt-brinjal moratorium boundary?

A. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. B. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. C. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. D. The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status.

Answer: D. Explanation: The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies GM-mustard legal-status boundary?

A. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. B. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. C. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. D. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes.

Answer: A. Explanation: The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of GM-mustard legal-status boundary?

A. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. B. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. C. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. D. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary.

Answer: B. Explanation: The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses GM-mustard legal-status boundary without changing its institution, unit or status?

A. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. B. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. C. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. D. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing.

Answer: C. Explanation: The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about GM-mustard legal-status boundary?

A. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. B. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. C. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. D. The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed.

Answer: D. Explanation: The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Genome-edited-rice status?

A. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. B. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. C. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. D. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary.

Answer: A. Explanation: ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Genome-edited-rice status?

A. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. B. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. C. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. D. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically.

Answer: B. Explanation: ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Genome-edited-rice status without changing its institution, unit or status?

A. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. B. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. C. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. D. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities.

Answer: C. Explanation: ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Genome-edited-rice status?

A. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. B. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. C. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. D. ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes.

Answer: D. Explanation: ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Resistance-and-refuge mechanism?

A. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. B. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. C. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. D. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing.

Answer: A. Explanation: Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Resistance-and-refuge mechanism?

A. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. B. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. C. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. D. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing.

Answer: B. Explanation: Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Resistance-and-refuge mechanism without changing its institution, unit or status?

A. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. B. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. C. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. D. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone.

Answer: C. Explanation: Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Resistance-and-refuge mechanism?

A. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. B. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. C. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. D. Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary.

Answer: D. Explanation: Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Gene-flow and non-target mechanism?

A. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. B. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. C. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. D. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing.

Answer: A. Explanation: Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Gene-flow and non-target mechanism?

A. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. B. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. C. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. D. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities.

Answer: B. Explanation: Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Gene-flow and non-target mechanism without changing its institution, unit or status?

A. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. B. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. C. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. D. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities.

Answer: C. Explanation: Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Gene-flow and non-target mechanism?

A. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. B. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. C. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. D. Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically.

Answer: D. Explanation: Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Off-target-characterisation boundary?

A. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. B. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. C. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. D. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities.

Answer: A. Explanation: CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Off-target-characterisation boundary?

A. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. B. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. C. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. D. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged.

Answer: B. Explanation: CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Off-target-characterisation boundary without changing its institution, unit or status?

A. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. B. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. C. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. D. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic.

Answer: C. Explanation: CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Off-target-characterisation boundary?

A. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. B. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. C. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. D. CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing.

Answer: D. Explanation: CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Farmer-livelihood boundary?

A. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. B. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. C. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. D. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities.

Answer: A. Explanation: Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Farmer-livelihood boundary?

A. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. B. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. C. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. D. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic.

Answer: B. Explanation: Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Farmer-livelihood boundary without changing its institution, unit or status?

A. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. B. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. C. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. D. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship.

Answer: C. Explanation: Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Farmer-livelihood boundary?

A. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. B. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. C. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. D. Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone.

Answer: D. Explanation: Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Approval-monitoring firewall?

A. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. B. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. C. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. D. Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic.

Answer: A. Explanation: Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Approval-monitoring firewall?

A. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. B. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. C. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. D. Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged.

Answer: B. Explanation: Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Approval-monitoring firewall without changing its institution, unit or status?

A. Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. B. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. C. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. D. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation.

Answer: C. Explanation: Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Approval-monitoring firewall?

A. DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. B. In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. C. A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. D. Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities.

Answer: D. Explanation: Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2018 GM-mustard and CRISPR objective demands, the 2019 artificial-chromosome, RNA-interference and Cas9 demands, and the 2019 GS-III farmer-livelihood question to this owner. Three representative cards preserve those routes; no objective answer key or option letter is supplied.

PYQ DEMAND CARD 1 - 2018 Prelims GS-I

Demand: Assess the routed statements on Indian GM mustard genes and properties together with CRISPR-Cas9 classification.

Status: Audited routed objective demands covering 2018 Q63 and the CRISPR portion of Q64; the official key is unavailable locally, so no option or answer letter is asserted.

Model solution: DMH-11 mechanism-status boundary: DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation.

CRISPR targeting boundary: In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without

a compatible PAM does not establish cutting. **GM-mustard legal-status boundary:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 1 - 2018 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: DMH-11 mechanism-status boundary: DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **CRISPR targeting boundary:** In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **GM-mustard legal-status boundary:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess the routed statements on Indian GM mustard genes and properties together with CRISPR-Cas9 classification. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Audited routed objective demands covering 2018 Q63 and the CRISPR portion of Q64; the official key is unavailable locally, so no option or answer letter is asserted. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: DMH-11 mechanism-status boundary: DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **CRISPR targeting boundary:** In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **GM-mustard legal-status boundary:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2018 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 2 - 2019 Prelims GS-I

Demand: Assess the routed concepts of artificial chromosomes, RNA interference and Cas9 function in molecular biology.

Status: Representative routed card covering 2019 Q93, Q96 and Q99; it distinguishes related molecular tools without asserting an objective answer key.

Model solution: CRISPR targeting boundary: In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **Repair-outcome boundary:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Base-prime editing boundary:** Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. **Approval-monitoring firewall:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 2 - 2019 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: CRISPR targeting boundary: In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **Repair-outcome boundary:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Base-prime editing boundary:** Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. **Approval-monitoring firewall:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

- 1. Claim and named evidence:** Demand: Assess the routed concepts of artificial chromosomes, RNA interference and Cas9 function in molecular biology. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
- 2. Claim and named evidence:** Status: Representative routed card covering 2019 Q93, Q96 and Q99; it distinguishes related molecular tools without asserting an objective answer key. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: CRISPR targeting boundary: In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **Repair-outcome boundary:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route

determines the edit. **Base-prime editing boundary:** Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. **Approval-monitoring firewall:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 2 - 2019 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 3 - 2019 GS-III

Demand: Discuss how biotechnology applications can improve farmers' living standards.

Status: Verified routed Mains demand, 15 marks and 250 words; the model route evaluates trait, field performance, costs, stewardship, markets and qualification rather than promising automatic gains.

Model solution: Bt-cotton mechanism: Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. **Resistance-and-refuge mechanism:** Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. **Farmer-livelihood boundary:** Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. **Approval-monitoring firewall:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on “PYQ DEMAND CARD 3 - 2019 GS-III”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Bt-cotton mechanism: Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. **Resistance-and-refuge mechanism:** Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. **Farmer-livelihood boundary:** Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. **Approval-monitoring firewall:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Discuss how biotechnology applications can improve farmers' living standards. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Status: Verified routed Mains demand, 15 marks and 250 words; the model route evaluates trait, field performance, costs, stewardship, markets and qualification rather than promising automatic gains. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Bt-cotton mechanism: Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. **Resistance-and-refuge mechanism:**

Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary.

Farmer-livelihood boundary: Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. **Approval-monitoring firewall:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "PYQ DEMAND CARD 3 - 2019 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish transgenic, cisgenic, intragenic and hybrid crops. Answer in about 150 words.

Model thesis: Claim: Engineering-breeding boundary. **Named evidence/example:** Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Transgenic-cisgenic-intragenic boundary. **Named evidence/example:** Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DMH-11 mechanism-status boundary. **Named evidence/example:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic.
- Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged.
- DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation.

Qualified conclusion: Claim: Engineering-breeding boundary. **Named evidence/example:** Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Transgenic-cisgenic-intragenic boundary. **Named evidence/example:** Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DMH-11 mechanism-status boundary. **Named evidence/example:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on "Distinguish transgenic, cisgenic, intragenic and hybrid crops. Answer in about 150 words.", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Engineering-breeding boundary. **Named evidence/example:** Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Transgenic-cisgenic-intragenic boundary. **Named evidence/example:** Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DMH-11 mechanism-status boundary. **Named evidence/example:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Engineering-breeding boundary. **Named evidence/example:** Genetic engineering deliberately changes genetic material through laboratory methods, whereas conventional hybrid breeding combines parental genomes through sexual crossing; a hybrid is not automatically genetically engineered, although DMH-11 is both a hybrid and transgenic. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Transgenic-cisgenic-intragenic boundary. **Named evidence/example:** Transgenic constructs use DNA from a sexually incompatible organism, cisgenic constructs transfer an intact gene from the same or a sexually compatible gene pool, and intragenic constructs rearrange regulatory or coding elements from that compatible gene pool; source and construct design must not be merged. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DMH-11 mechanism-status boundary. **Named evidence/example:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Distinguish transgenic, cisgenic, intragenic and hybrid crops. Answer in about 150 words.", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain CRISPR-Cas9 targeting and the role of guide RNA, Cas9 and PAM. Answer in about 150 words.

Model thesis: **Claim:** CRISPR targeting boundary. **Named evidence/example:** In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Repair-outcome boundary. **Named evidence/example:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting.
- A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit.

Qualified conclusion: **Claim:** CRISPR targeting boundary. **Named evidence/example:** In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Repair-outcome boundary. **Named evidence/example:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain CRISPR-Cas9 targeting and the role of guide RNA, Cas9 and PAM. Answer in about 150...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** CRISPR targeting boundary. **Named evidence/example:** In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Repair-outcome boundary. **Named evidence/example:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: CRISPR targeting boundary. **Named evidence/example:** In CRISPR-Cas9 editing, a guide RNA provides sequence recognition, Cas9 is the nuclease, and cleavage requires an adjacent PAM such as NGG for commonly used SpCas9; guide matching without a compatible PAM does not establish cutting. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Repair-outcome boundary. **Named evidence/example:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain CRISPR-Cas9 targeting and the role of guide RNA, Cas9 and PAM. Answer in about 150...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Compare NHEJ, HDR, base editing, prime editing and SDN categories. Answer in about 250 words.

Model thesis: Claim: Repair-outcome boundary. **Named evidence/example:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Base-prime editing boundary. **Named evidence/example:** Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SDN-category boundary. **Named evidence/example:** SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a

targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** 2022 exemption-guideline boundary. **Named evidence/example:** MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit.
- Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ.
- SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption.
- MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks.

Qualified conclusion: **Claim:** Repair-outcome boundary. **Named evidence/example:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Base-prime editing boundary. **Named evidence/example:** Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SDN-category boundary. **Named evidence/example:** SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** 2022 exemption-guideline boundary. **Named evidence/example:** MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **compare** requires a direct position on "Compare NHEJ, HDR, base editing, prime editing and SDN categories. Answer in about 250 words.", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Repair-outcome boundary. **Named evidence/example:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Base-prime editing boundary. **Named evidence/example:** Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SDN-category boundary. **Named evidence/example:** SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** 2022 exemption-guideline boundary. **Named evidence/example:** MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the edit. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** Repair-outcome boundary. **Named evidence/example:** A Cas9 double-strand break is repaired mainly by error-prone non-homologous end joining, which creates small insertions or deletions, or by homology-directed repair using a donor template for a defined substitution or insertion; the repair route determines the

edit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Base-prime editing boundary. **Named evidence/example:** Base editing chemically converts a target base without a double-strand break, while prime editing uses a reverse-transcriptase-linked editor and prime-editing guide RNA to write a specified short change; neither should be described as ordinary NHEJ. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SDN-category boundary. **Named evidence/example:** SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** 2022 exemption-guideline boundary. **Named evidence/example:** MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Compare NHEJ, HDR, base editing, prime editing and SDN categories. Answer in about 250 words.", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Map India's six-tier biosafety architecture and separate GEAC, RCGM and FSSAI roles. Answer in about 250 words.

Model thesis: **Claim:** Six-tier biosafety architecture. **Named evidence/example:** The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GEAC-RCGM-FSSAI split. **Named evidence/example:** RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rules-1989 boundary. **Named evidence/example:** The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental

release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels.
- RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others.
- The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate.
- Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities.

Qualified conclusion: Claim: Six-tier biosafety architecture. **Named evidence/example:** The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GEAC-RCGM-FSSAI split.

Named evidence/example: RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rules-1989 boundary. **Named evidence/example:** The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall.

Named evidence/example: Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on "Map India's six-tier biosafety architecture and separate GEAC, RCGM and FSSAI roles. Answer...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Six-tier biosafety architecture. **Named evidence/example:** The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GEAC-RCGM-FSSAI split.

Named evidence/example: RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rules-1989 boundary. **Named evidence/example:** The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** Six-tier biosafety architecture. **Named evidence/example:** The Rules, 1989 create a six-tier structure: RDAC advises, IBSC oversees work within institutions, RCGM reviews research and contained activity, GEAC appraises environmental release and large-scale use, and SBCC plus DLC monitor at state and district levels. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GEAC-RCGM-FSSAI split. **Named evidence/example:** RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. **Analysis:** This fixes the scientific process, system boundary,

institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rules-1989 boundary. **Named evidence/example:** The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Map India's six-tier biosafety architecture and separate GEAC, RCGM and FSSAI roles. Answer...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Critically examine India's regulation of GM crops and genome-edited plants after the 2022 exemption. Answer in about 300 words.

Model thesis: **Claim:** SDN-category boundary. **Named evidence/example:** SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GEAC-RCGM-FSSAI split. **Named evidence/example:** RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rules-1989 boundary. **Named evidence/example:** The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** 2022 exemption-guideline boundary. **Named evidence/example:** MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Bt-brinjal moratorium boundary. **Named evidence/example:** The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GM-mustard legal-status boundary. **Named evidence/example:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption.
- RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others.
- The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate.
- MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks.
- The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status.
- The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed.
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Qualified conclusion: **Claim:** SDN-category boundary. **Named evidence/example:** SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GEAC-RCGM-FSSAI split. **Named evidence/example:** RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rules-1989 boundary. **Named evidence/example:** The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before

drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** 2022 exemption-guideline boundary. **Named evidence/example:** MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Bt-brinjal moratorium boundary. **Named evidence/example:** The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GM-mustard legal-status boundary. **Named evidence/example:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **critically examine** requires a direct position on “Critically examine India's regulation of GM crops and genome-edited plants after the 2022...”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** SDN-category boundary. **Named evidence/example:** SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GEAC-RCGM-FSSAI split. **Named evidence/example:** RCGM under DBT addresses research and contained stages, GEAC under MoEF&CC appraises environmental release, and FSSAI addresses food-safety assessment, approval and labelling under the food-law track; no one body substitutes for the others. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rules-1989 boundary. **Named evidence/example:** The Manufacture, Use, Import, Export and Storage of Hazardous Micro-organisms, Genetically Engineered Organisms or Cells Rules, 1989 under the Environment Protection Act, 1986 are India's core GMO biosafety framework, not a crop-commercialisation certificate. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** 2022 exemption-guideline boundary. **Named evidence/example:** MoEF&CC's 30 March 2022 office memorandum exempted SDN-1 and SDN-2 genome-edited plants free of exogenously introduced DNA from Rules 7 to 11, while DBT's 17 May 2022 safety-assessment guidelines and related oversight still preserve category and biosafety checks. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Bt-brinjal moratorium boundary. **Named**

evidence/example: The 9 February 2010 Bt brinjal decision was a moratorium pending further independent long-term safety evidence, not a permanent statutory ban, commercial approval or proof that every later GM-food proposal has the same status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GM-mustard legal-status boundary. **Named evidence/example:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** SDN-1 makes a template-free small edit, SDN-2 uses a short homologous template for a defined small change, and SDN-3 inserts a gene-sized sequence at a targeted site; SDN-3 DNA may be cisgenic or transgenic and remains outside India's lighter exemption. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
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7. **Claim and named evidence:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Critically examine India's regulation of GM crops and genome-edited plants after the 2022...”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Evaluate agricultural biotechnology through Bt cotton, DMH-11, genome-edited rice, biosafety and farmer livelihoods. Answer in about 300 words.

Model thesis: Claim: Bt-cotton mechanism. **Named evidence/example:** Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DMH-11 mechanism-status boundary. **Named evidence/example:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GM-mustard legal-status boundary. **Named evidence/example:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Genome-edited-rice status. **Named evidence/example:** ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Resistance-and-refuge mechanism. **Named evidence/example:** Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Gene-flow and non-target mechanism. **Named evidence/example:** Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Off-target-characterisation boundary. **Named evidence/example:** CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Farmer-livelihood boundary. **Named evidence/example:** Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system

category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship.
- DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation.
- The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed.
- ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes.
- Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary.
- Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically.
- CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing.
- Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone.
- Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities.

Qualified conclusion: Claim: Bt-cotton mechanism. **Named evidence/example:** Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DMH-11 mechanism-status boundary. **Named evidence/example:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GM-mustard legal-status boundary. **Named evidence/example:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Claim: DMH-11 mechanism-status boundary. **Named evidence/example:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GM-mustard legal-status boundary. **Named evidence/example:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

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Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Genome-edited-rice status. **Named evidence/example:** ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Resistance-and-refuge mechanism. **Named evidence/example:** Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Gene-flow and non-target mechanism. **Named evidence/example:** Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Off-target-characterisation boundary. **Named evidence/example:** CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Farmer-livelihood boundary. **Named evidence/example:** Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **evaluate** requires a direct position on “Evaluate agricultural biotechnology through Bt cotton, DMH-11, genome-edited rice, biosafety...”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Bt-cotton mechanism. **Named evidence/example:** Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DMH-11 mechanism-status boundary. **Named evidence/example:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GM-mustard legal-status boundary. **Named evidence/example:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges

differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Genome-edited-rice status. **Named evidence/example:** ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Resistance-and-refuge mechanism. **Named evidence/example:** Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Gene-flow and non-target mechanism. **Named evidence/example:** Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Off-target-characterisation boundary. **Named evidence/example:** CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Farmer-livelihood boundary. **Named evidence/example:** Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an

appropriate Bench; commercial cultivation and settled legality must not be claimed. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
9. **Claim and named evidence:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Bt-cotton mechanism. **Named evidence/example:** Bt cotton expresses cry proteins derived from *Bacillus thuringiensis* against specified lepidopteran bollworms; insect resistance is distinct from herbicide tolerance, and stacked cry genes delay but do not eliminate resistance evolution or refuge stewardship. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DMH-11 mechanism-status boundary. **Named evidence/example:** DMH-11 uses the barnase-barstar male-sterility and fertility-restoration system, with bar as a selectable marker, to enable mustard hybridisation; reviewed official material concerns conditional environmental release for seed production and testing, not established commercial cultivation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GM-mustard legal-status boundary. **Named**

evidence/example: The Supreme Court delivered a split verdict on 23 July 2024 concerning DMH-11: the judges differed on the challenged approvals and the divided issue was placed before the Chief Justice for an appropriate Bench; commercial cultivation and settled legality must not be claimed. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Genome-edited-rice status. **Named evidence/example:** ICAR announced DRR Rice 100 Kamla and Pusa DST Rice 1 on 4 April 2025 as genome-edited rice varieties developed without adding foreign DNA; announcement of varieties does not establish seed availability, cultivated area or farmer outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Resistance-and-refuge mechanism. **Named evidence/example:** Continuous selection pressure can increase resistant pest alleles, so structured refuges preserve susceptible insects and gene stacking can slow resistance; Bt technology does not make resistance management unnecessary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Gene-flow and non-target mechanism. **Named evidence/example:** Pollen or seed movement can transfer traits to compatible relatives or neighbouring crops, while exposure pathways may affect non-target organisms; risk assessment must examine crop biology, receiving environment and trait rather than treating every event identically. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Off-target-characterisation boundary. **Named evidence/example:** CRISPR off-target changes arise when a nuclease acts at sufficiently similar sequences; guide design, high-fidelity enzymes, segregation and sequencing can reduce or identify them, but targeted editing is not synonymous with consequence-free editing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Farmer-livelihood boundary. **Named evidence/example:** Biotechnology can affect yield stability, pesticide exposure, labour, seed cost, resistance management and market access, so farmer welfare depends on trait performance, stewardship, extension, competition and local agro-ecology rather than the technique label alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-monitoring firewall. **Named evidence/example:** Laboratory result, contained research permission, confined field trial, environmental release, food approval, crop-variety release, seed availability, commercial cultivation and post-release monitoring are separate evidence rungs that require their own dated authorities. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Evaluate agricultural biotechnology through Bt cotton, DMH-11, genome-edited rice, biosafety...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.